Mission: Smart Warehouse Core API

Version 2.0 · 28 October 2026 · Replaces the earlier brief (shelf-locking and queue design)

Objective

Build the backend for the Smart Warehouse WMS: a .NET Web API (controller, service and repository layers, EF Core, SQL Server) serving the Android worker app, the ASP.NET MVC admin dashboard (which calls the API like any other client) and the ESP32 LED controllers. The full schema and endpoint list are in the ER Schema and API Specification v2.0. This brief says what to build, in what order, and how to know it works.

Rules that must hold

1. No shelf locking, no reservation queue and no waiting. Orders run in parallel.
2. Availability = OnHand − Committed, per shelf. New orders are validated against Available, never raw OnHand.
3. Order creation validates all lines first, then writes all lines in one transaction using conditional updates. Any failure rolls back everything and returns 409 naming the short shelf.
4. A confirmed pick lowers OnHand and Committed together, in one transaction.
5. Cancel is allowed only before the first confirmed pick, and it releases all Committed stock for the order.
6. The admin chooses the shelf and the worker for every order line. There is no automatic assignment or load balancing.
7. One product per shelf. One admin per warehouse.
8. Handoff is confirmed per worker, and the order completes when every line is handed off. Handoff does not touch stock.

Build order

1. Entities, EF Core mappings, migration and seed data: products, product types, shelves, a small map graph, one admin and two workers.
2. Authentication and roles (Admin, Worker).

3. Read endpoints for shelves and inventory, with computed Available and empty and full flags.
4. Order creation with a worker on every line, cancel, and line reassignment (the transaction logic).
5. Worker endpoints: my orders, confirm pick, confirm handoff.
6. Inbound: create, assign shelf, confirm placement.
7. Map and route endpoints (Dijkstra over the in-memory graph).
8. Device endpoint that tells each ESP32 which LEDs to light.

Acceptance checks

- Two orders on the same shelf whose combined cartons exceed Available: the second is rejected with 409 and nothing from it stays committed.
- A three-line order whose third line is short leaves Committed unchanged on the first two shelves.
- After a pick, Available on that shelf is unchanged while OnHand and Committed each drop by the picked cartons.
- Cancel after the first pick returns 409. Cancel before it restores Committed to its earlier value.
- A worker who is not assigned to a line gets 403 when picking it, and a route covers only the caller's own lines.
- The route from any shelf anchor visits the remaining shelves in rack-then-level order and ends at the dock.
- Placing inbound stock on an empty shelf sets the shelf's product, and emptying the shelf clears it.

Out of scope

Branch self-submission, SignalR (the dashboard polls), automatic worker assignment, reordering stops to shorten routes, a map editor, and live positioning (the worker names the shelf they are standing next to).

Open items

None blocking. Handoff per worker per order is confirmed. The map graph will be seeded from the exported floor-map image before implementation starts.